

Contents

Adaptive evidence integration as an information bottleneck	1
Abstract	1
Orientation	2
A conceptual guide to the IB setup	2
Part I — The central result: capacity quantizes the belief	4
1. The bottleneck does not find a new variable; it coarsens the normative one	4
2. What “coarsening” turns out to mean: a staircase, not a blur	6
2.1 Why quantization rather than noise	7
2.2 Where the boundaries go, and why the branches are uneven	10
2.3 The first transition has a closed form	10
2.4 The empirical prediction	11
Part II — Robustness of the quantization result	11
3. Continuous-valued observations do not dissolve it	11
4. The window W is a numerical device, not a capacity ceiling	14
5. Hazard dependence, including where W does bind	14
β as a price per bit of capacity	14
Part III — Recursion	16
6. Requiring the belief to be updated from itself is costly, and costly exactly where subjects are	16
7. The recursive transition is abrupt	18
8. Methods	19
9. What this buys the grant	19
10. Open items	20
11. Files	20

Adaptive evidence integration as an information bottleneck

Working memo v5, annotated. All numbers are computed from the current computational core (`ib_core.py`); §8 gives the methods.

Abstract

This memo applies the information bottleneck (IB) framework to the problem of adaptive evidence integration in a volatile environment, using the “beads task” in which an observer tracks a latent source that switches unpredictably over time. The central result is that capacity constraints do not blur the normative belief (the Glaze log-posterior-odds variable) but instead quantize it: the IB-optimal representation partitions the belief axis into a small number of contiguous discrete categories rather than producing a noisy, graded continuous belief. This categorical structure arises because the IB objective rewards distinctions, not precision, and is proven to be a geometric necessity given binary KL divergence. The first transition from “ignore evidence” to “hold a binary belief” has a closed-form threshold and is continuous (second-order). In contrast, the analogous transition under recursive inference is abrupt (first-order) because of a non-convexity in the achievable region. For the resulting IB curve, SNR sets the evidence ceiling, whereas capacity alone determines the number of belief categories. Computing the bound on-line instead of after experiencing all the observations, which adds a major cost in terms of a loss in predictiveness, is basically a discrete version of the normative (Bayesian) dynamics, involving a leaky accumulator with non-absorbing bounds.

Orientation

The bead task asks a subject to watch a stream of coloured beads drawn from one of two jars, where the jar occasionally switches without warning, and to report which jar is the current source. A good observer accumulates evidence across beads but discounts old evidence, because the jar may have changed. Glaze et al. (2015) derived the normative solution: a log-posterior-odds variable updated by

$$L_t = \psi(L_{t-1}) + \text{LLR}(x_t), \quad \psi(L) = L + \log[(1-h)+h \cdot e^{-L}] - \log[(1-h)+h \cdot e^{L}]$$

where h is the hazard rate. ψ discounts old evidence, and its effect is to bound L at a *stabilizing* amplitude $\pm \log[(1-h)/h]$ — the point past which further evidence in the same direction stops moving the belief.

This memo asks what happens when such an observer is **limited in how much it can retain** about the beads it has seen. The tool is the information bottleneck (IB): for each budget of retained information, find the strategy that best predicts the current jar.

A conceptual guide to the IB setup

Before presenting the results, it is worth being explicit about several aspects of the IB framework that are easy to misread.

What is R , and what form does it take?

R is the observer's internal representation — whatever compressed summary of the bead sequence the observer retains. The IB optimization places almost no constraints on R going in: it could be continuous, discrete, high-dimensional, or any other form. No structure is assumed. What determines R 's form is entirely the optimization itself.

For finite X (e.g., the 128 distinct bead sequences in a $W=7$ window), a classical result from convex geometry (Carathéodory's theorem) guarantees that any point on the IB bound achievable by some continuous R is also achievable by a discrete R with at most $|X|$ states. Restricting to discrete R is therefore without loss of generality — nothing is thrown away. The Blahut-Arimoto algorithm exploits this: it optimizes over a $k \times |X|$ matrix of assignment probabilities $p(r|x)$, treating the k states of R as a free parameter.

Even though up to 128 states are formally available, the optimization finds that only a handful earn their keep. The reason is that the evidence ceiling $I(X;Y)$ — the most any observer could know about the jar — is small (≈ 0.806 bits at $h=0.01$). Most states would carry no additional information about the jar even if used.

The connection to the “add noise to the posterior” result

A classical result in IB theory states that the IB-optimal encoder depends on x only through the posterior over the target: $p(r|x)$ is a function of $q(x) = p(Y=1|x)$. Put plainly: R should be a noisy function of the posterior over the latent state.

A note on notation that is easy to trip over: $q(x)$ is written as a function of x (the observation sequence), but it is a probability over Y (the latent jar). It is exactly the posterior over the latent state — the same quantity as “the posterior” in Parker et al. eq. 5. The two notations emphasize different things: $q(x)$ stresses that the posterior varies across observations; “the posterior over Y ” stresses what it is a belief about. They are the same object.

The “noise” in the “add noise to posterior” result lives in the **assignment to R** , not in $q(x)$ itself. Given a posterior value $q(x)$, the observer does not deterministically land in one state of R — there is residual randomness about which state gets assigned. This is the noise. It is categorically different from adding Gaussian noise to L_t (which would smear the belief value itself). The output R is still a small set of ordered categories; what is stochastic is which category a given observation lands in, especially near the boundaries between categories. As β increases, those boundaries sharpen into hard thresholds.

Why the optimal R is discrete rather than a blurred continuous variable

The intuitive default is that a capacity limit should make the belief *noisy* — the same graded, continuous confidence, only corrupted. This is not what the IB produces, for a fundamental reason: **the IB objective pays for distinctions, not for precision.**

Adding Gaussian noise to a continuous belief pays the full capacity cost of a high-resolution variable (high-entropy R) while realizing only part of the benefit. Sorting the belief axis into a few well-separated categories pays for exactly the distinctions it uses. The optimum is therefore atomic — a small set of discrete states — and the “noise” from the classical result manifests as soft, probabilistic boundaries between those states, not as a blur of the belief value itself.

The reconciliation with Parker et al. eq. 5 is then as follows. Both results derive from the same IB fixed-point equation. The classical result correctly says R is a noisy function of $q(x)$. This memo adds the structural question: *what form does that noisy function take when R is low-capacity?* The answer is an interval partition with soft boundaries. The two framings are the same object viewed differently:

	Variable that is noisy	What R looks like
Intuitive “blur” model	L_t itself	Continuous, smeared belief
IB “add noise” result	Assignment r given $q(x)$	Discrete state; stochastic at boundaries

These have different behavioral predictions. The blur model predicts graded confidence varying smoothly with L_t . The IB model predicts categorical confidence with a sharp (if noisy) criterion — and what varies with capacity is how sharp that criterion is, not how graded the belief is.

symbol	name	units	meaning
h	hazard rate	—	probability the source switches between trials
d'	discriminability	—	signal-to-noise of one observation
L_t	belief	nats	log posterior odds; the Glaze decision variable
$\log[(1-h)/h]$	stabilizing bound	nats	amplitude at which L stops growing
X	evidence	—	the bead sequence, or equivalently L
Y	target	—	the current source jar
R	representation	—	the observer’s internal state
$I(X;R)$	capacity	bits	what the observer retains about the evidence
$I(R;Y)$	predictiveness	bits	what that buys in jar-knowledge — the payoff
$I(X;Y)$	evidence ceiling	bits	the most any observer could achieve
β	trade-off	—	price paid per bit of capacity
β_c	critical β	—	below it, the optimum ignores the evidence

symbol	name	units	meaning
k	states	—	number of distinguishable values R takes

Two cautions on quantities that are easy to conflate. First, $I(X;Y)$ and $\log[(1-h)/h]$ are different objects with different units — a mutual information in bits versus an amplitude on the belief axis in nats — and they are not related by any simple transform. At $h = 0.01$ they are 0.806 bits and 4.595 nats; across $h \in [0.002, 0.3]$ the bound falls $7.3\times$ while $I(X;Y)$ falls $3.0\times$, and as $h \rightarrow 0.5$ the bound $\rightarrow 0$ while $I(X;Y) \rightarrow 0.278$ and stays there. Second, **every number in this memo is hazard-dependent**; each table states its h , and §5 gives the hazard dependence explicitly, since it changes one conclusion materially.

Terms coined here (not standard, and not findable in the literature): *bounded ladder* — an observer holding one of a few ordered belief levels, moved up or down one level per bead, with the extreme levels non-absorbing (the same object Glaze et al. call a leaky accumulator with non-absorbing bounds); *cost of recursion* — the drop in predictiveness from requiring the observer to update its state from its own previous state rather than recomputing from the window afresh, at matched capacity.

Part I — The central result: capacity quantizes the belief

1. The bottleneck does not find a new variable; it coarsens the normative one

Solve the IB on the bead task with no assumption that the answer resembles the normative model, then ask afterward what the solution turned out to be a function of. Two facts answer that, and both are exact rather than statistical.

Figure 1. The bottleneck coarsens the normative belief, and coarsens it into intervals.

(a) The HMM forward posterior $p(y_t = 1|x)$ — computed by exact recursion over the hidden Markov chain, with no reference to Glaze — against $\sigma(L_t)$, the logistic transform of the belief produced by the Glaze recursion on the same sequence. All 128 length-7 windows, $W = 7$, $h = 0.01$; maximum deviation 2.2×10^{-16} . The Glaze recursion is an exact online implementation of the forward algorithm, so the bottleneck’s only channel to the stimulus is the normative belief itself. This panel establishes that $q(x) = p(Y=1|x)$ — the posterior over the latent jar — is the sufficient statistic: the IB has nothing to work with beyond what Glaze already computes.

(b) Why the assignment must be an interval partition. Under the IB encoder the r -dependent part of the exponent is affine in $q(x) = p(y_t = 1|x)$, so each state’s assignment score is a straight line in q whose slope is that state’s decoder log-odds. To see why: at any stationary point of the IB objective, state r ’s score for window x is proportional to $\exp(-\beta \cdot D_{\text{KL}}[p(y|x) \parallel p(y|r)])$. Expanding the KL divergence for binary Y and collecting terms that depend on both x and r gives a quantity that is linear (affine) in $q(x)$, with slope equal to state r ’s decoder log-odds $\log(q_r/(1-q_r))$. The winning state at each q is the upper envelope of those lines (grey). An upper envelope of straight lines necessarily partitions the axis into contiguous intervals ordered by slope — a state cannot win at two separated regions of q without its line crossing above all others twice, which is impossible for a straight line. Four lines are shown, corresponding to the four states of (c). This is a proof by geometry: the interval partition structure is not a choice or assumption — it is forced by the algebra of KL divergence applied to a binary target.

(c) The $\beta = 9$ solution: stacked assignment probabilities $p(r|x)$ for all 128 windows against L_t , colours matching (b), with the interval boundaries at $L = -1.49, -0.27, +1.43$ nats (dashed). Each colour dominates a contiguous stretch of the L axis. The soft transitions between colours at finite β are the “noise” from the classical IB result — near a boundary, there is genuine uncertainty about which state a window is assigned

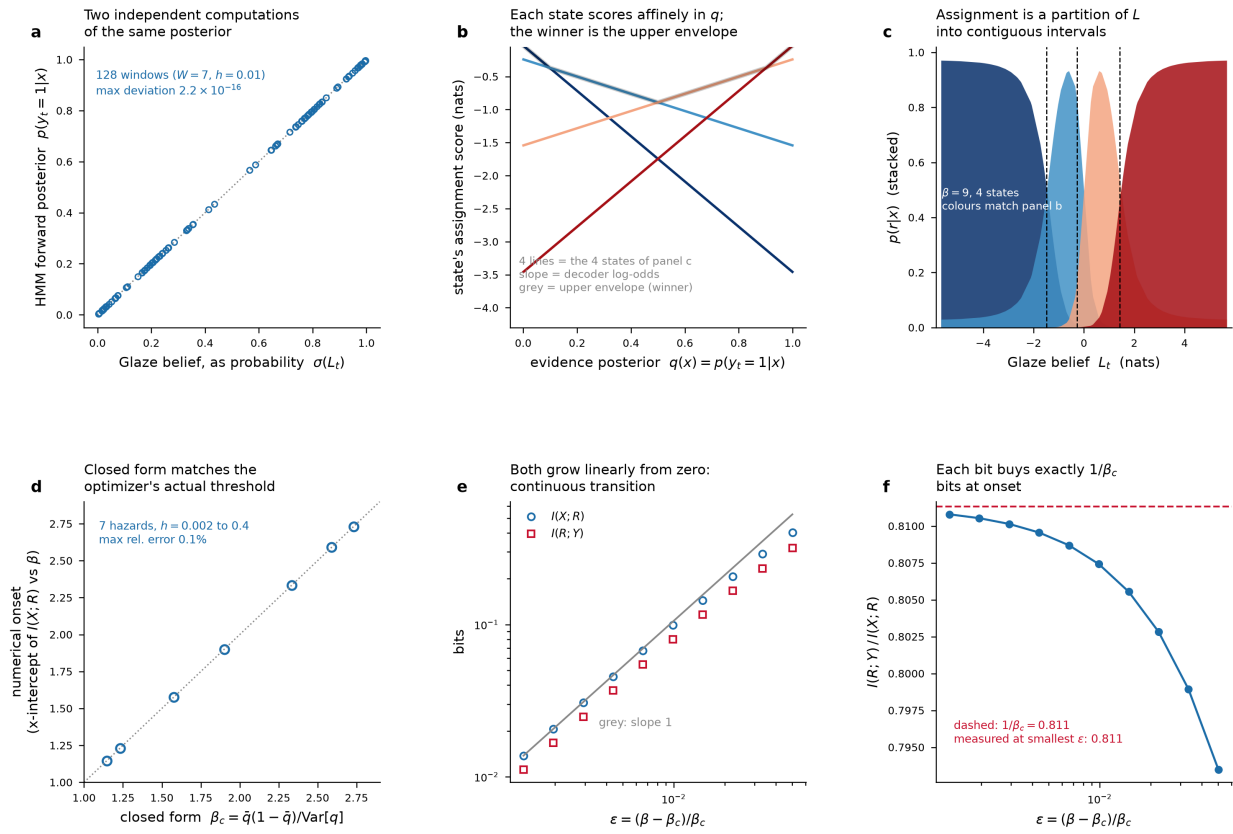


Figure 1: The bottleneck coarsens the normative belief into contiguous intervals

to. As β increases, these soft transitions sharpen into hard thresholds. The observer holds whichever state's decoder value corresponds to their assigned state — not a smeared version of L_t .

(d) The closed-form threshold $\beta_c = \bar{q}(1 - \bar{q})/\text{Var}[q]$ against the numerical onset recovered from the optimizer, over 7 hazards from $h = 0.002$ to 0.4 ; maximum relative error 0.1% . β_c is the minimum exchange rate between predictiveness and capacity at which drawing even the first boundary pays off. The numerator $\bar{q}(1-\bar{q})$ measures prior uncertainty about the jar on average; the denominator $\text{Var}[q]$ measures how much the evidence moves the posterior around across different observation sequences. Their ratio is the break-even price: when evidence barely moves q (small $\text{Var}[q]$), β_c is large and the observer needs to value predictiveness highly before encoding anything pays off. All 7 points fall on the diagonal, validating the formula across a wide range of hazard rates. See §2.3.

(e) $I(X;R)$ and $I(R;Y)$ just above threshold, both growing linearly in $\varepsilon = (\beta - \beta_c)/\beta_c$ (grey line, slope 1): the transition is continuous. ε is the reduced distance from the critical point — standard notation from statistical physics, analogous to reduced temperature $t = (T-T_c)/T_c$ near a phase transition. Both quantities grow as ε^1 (slope 1 on a log-log plot), the defining signature of a continuous (second-order) phase transition. The observer smoothly transitions from “ignores evidence entirely” to “holds a barely-distinguishable binary belief” as β crosses β_c — there is no discontinuous jump. The exponent of exactly 1 follows from the perturbation argument in §2.3: both the capacity cost and predictiveness gain of a small split scale as δ^2 , so they cancel and growth near threshold is exactly linear. Contrast with the recursive transition (§7), which is abrupt (first-order).

(f) Their ratio $I(R;Y)/I(X;R)$ extrapolates to $1/\beta_c = 0.811$ as $\varepsilon \rightarrow 0$ — at onset each bit of capacity buys exactly $1/\beta_c$ bits of predictiveness. This is the break-even exchange rate at the moment of onset: β_c is the price at which retaining information first becomes worthwhile, so $1/\beta_c$ is the return on the first bit invested. As ε increases (moving further above threshold), the ratio drops — later bits buy less predictiveness because the most informative boundary (near $q=0.5$) is already in use, and additional capacity goes toward sharpening it or drawing new boundaries in less informative regions of the belief axis.

The bottleneck therefore does not discover a new variable. It depends on the stimulus only through $p(y|x)$, which is $\sigma(L_{\text{Glaze}})$ to machine precision (a), and what it does to that quantity is cut it into contiguous intervals (b, c). So the capacity limit does not change **what** is represented. It changes only the **resolution** at which the normative belief is held. Everything that follows is a statement about that resolution.

2. What “coarsening” turns out to mean: a staircase, not a blur

This is the main result of the memo, and it is worth stating slowly because the intuitive default is wrong.

The intuitive default is that a capacity limit should make the belief **noisy** — that a subject with half the information capacity holds the same graded, continuous confidence, only corrupted. That is not what the optimization does. Instead, at any capacity below the evidence ceiling, the optimal representation **partitions the belief axis into a small number of contiguous intervals** and retains only which interval the belief is in. The observer holds a *category*, not a blurred number.

Figure 2. Capacity quantizes the normative belief.

(a) IB bound for the bead task ($h = 0.01$, $w = 7$, $Y = \text{current jar}$). The x-axis is information capacity $I(X;R)$ — how many bits the observer retains about the evidence. The y-axis is predictiveness $I(R;Y)$ — how many bits that buys about which jar is active. Every point on the curve is the best possible predictiveness achievable at that capacity — no observer can be above the curve. Colour gives the number of distinct belief states k in the optimal solution; the curve is a sequence of discrete branches, each holding a fixed number of states over a whole range of capacities. This staircase shape is the central result: the frontier is not smooth because the optimizer finds it better to sharpen existing boundaries before adding new ones. Within each branch, extra capacity makes the assignment more deterministic (sharper thresholds) without creating new categories. New states appear only when sharpening is exhausted, producing a discrete jump to the next branch. Dotted line is the evidence ceiling $I(X;Y) = 0.76$ bits. Note that the curve flattens quickly — most available predictiveness is captured by just 2–4 states.

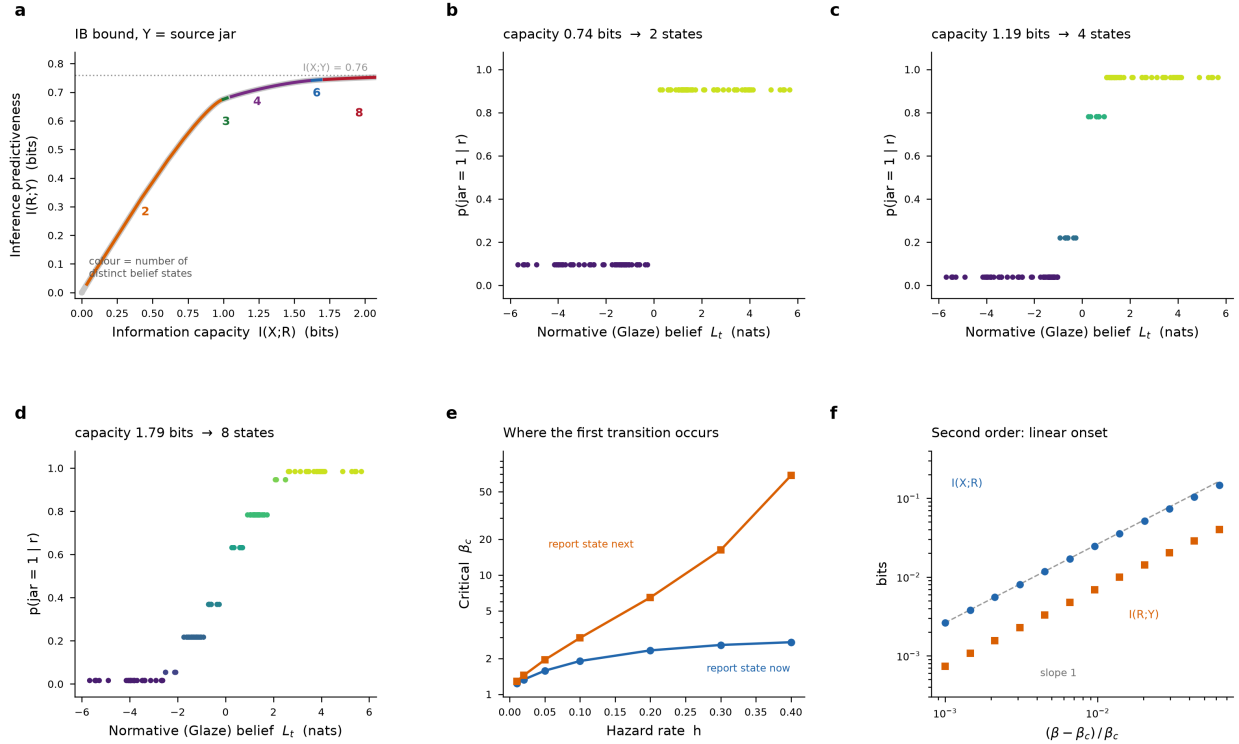


Figure 2: Quantization of the normative belief by the information bottleneck

(b–d) The optimal decoder $p(\text{jar} = 1 \mid r)$ plotted against the normative belief L_t , at three capacities. Each panel is a **step function of L** : the optimum assigns an interval of the belief axis to each state, with each horizontal line segment showing one state’s decoder value over the range of L values assigned to it. At 0.74 bits there are two states split near $L = 0$ (b): a left state (“probably jar 2”, decoder ≈ 0.06) and a right state (“probably jar 1”, decoder ≈ 0.94) — a binary belief. At 1.19 bits, four states (c): extreme states remain near 0 and 1; two new middle states appear near 0.2 and 0.8. At 1.79 bits, eight states (d): further carving of the middle, while tails remain covered by single states. Increasing capacity changes only how finely the L axis is cut — the observer always holds a category. Boundaries cluster within roughly $|L| < 2$ because the optimizer places them where $(\text{time spent}) \times (\text{usefulness of resolving there})$ is maximised (§2.2).

(e) Location of the first transition, β_c , versus hazard rate h , for two task instructions (predict current jar vs. predict next jar). β_c rises with h because faster switching makes the evidence less useful — the denominator $\text{Var}[q]$ falls as the belief is less reliably moved by evidence when the source switches frequently. The gap between the two instructions widens dramatically at high h : $\beta_c^{\text{next}} = \beta_c^{\text{now}} / (1-2h)^2$, which diverges as $h \rightarrow 0.5$. At $h=0.40$, $\beta_c^{\text{next}} \approx 68.3$ while $\beta_c^{\text{now}} \approx 1.9$ — a capacity-limited observer should rationally disengage from prediction much sooner than from filtering as switching gets frequent. This is a parameter-free prediction about instruction effects, testable by changing only the question asked at the end of each trial.

(f) Growth of capacity and predictiveness just above β_c : both are linear in $(\beta - \beta_c) / \beta_c$, the signature of a continuous (second-order) transition. Shown here to confirm it holds across the range of hazard rates in panel (e), not just at $h=0.01$.

2.1 Why quantization rather than noise

Three facts, in order.

First: a bit is a distinction. The capacity $I(X;R)$ measures how many distinguishable states the represen-

tation carries. This is not a metaphor — a representation carrying one bit can support exactly two reliably distinguishable states. Since Y is binary, everything the observer needs from its state is a single number, $q_r = p(jar = 1 | r)$. So the “codebook” the observer must build lives on a one-dimensional interval, and a capacity budget is directly a budget on how many points it may place in that interval.

Second: the optimization pays for distinctions, not for precision. At any stationary point of the IB objective the encoder must satisfy

$$p(r|x) \propto p(r) \cdot \exp(-\beta \cdot D_{KL}[p(y|x) \parallel p(y|r)])$$

This is not an assumption; it is what setting the derivative of $I(X;R) - \beta I(R;Y)$ to zero gives, and it is the update Blahut–Arimoto iterates. Read it as a scoring rule: each candidate state r scores window x by how well r ’s decoder $p(y|r)$ matches x ’s true posterior $p(y|x)$, and β sets how decisively the best-matching state wins. The cost term $I(X;R)$ charges for *how reliably* x determines r ; the benefit term charges for *how different* the decoders are. Neither term rewards spreading mass smoothly over the belief axis. Adding Gaussian-like noise to a continuous belief pays the full capacity cost of a high-resolution variable while realizing only part of the benefit; sorting the axis into a few well-separated categories pays for exactly the distinctions it uses. The optimum is therefore atomic.

The reconciliation with the classical “add noise to the posterior” result (Parker et al. eq. 5; Tishby et al. 1999) follows directly. The classical result correctly says R is a noisy function of $q(x)$. This memo adds the structural question: *what form does that noisy function take when R is low-capacity?* The answer is an interval partition with soft boundaries. The “noise” is not a blur of $q(x)$ — it is uncertainty about which discrete category a given $q(x)$ maps to, concentrated at the boundaries between categories.

Third: capacity buys two separable things, and it buys them in order. Within a branch, extra capacity is spent making the assignment **sharper** at a fixed number of states, and only when sharpening is exhausted does a new state appear.

Figure 3. Sharpen, then split.

(a) The two decoders pull apart. Within the two-state branch, the decoder values $q_r = p(jar = 1 | r)$ of the upper and lower state against capacity $I(X;R)$. At the bottom of the branch they sit at 0.61 and 0.39 — the two states say almost the same thing — and by the top of the branch at 0.94 and 0.06. The shaded region is the gap between them.

(b) Assignment goes from coin-flip to deterministic. Mean assignment entropy $H[p(r|x)]$ against capacity. It falls from 0.957 bits (each window split almost 50/50 between the two states) to 0.012 bits (each window assigned to one state). Nothing about the *number* of states changes across this range. “Sharpness” here is a specific quantity — the mean assignment entropy — not an impression.

(c) The criterion stiffens. $p(\text{upper state}|x)$ against the Glaze belief L_t at three capacities (0.30, 0.86, 0.99 bits). The crossing stays at $L = 0$; what changes is the steepness — a soft criterion becomes a hard threshold. The boundary location is fixed from the start; what capacity buys is a sharper transition.

(d) Higher branches leave alphabet capacity unused. Capacity actually reached at the top of each k -state branch, against the alphabet ceiling $\log_2 k$ (grey dashed), for $k = 2$ to 14. Only $k = 2$ runs to saturation (0.992 of $\log_2 2$); every branch above it tops out well short.

(e) Why: the middle states are rarely occupied. Occupancy $p(r)$ of the four states at $\beta = 9$, against each state’s decoder q_r . The two extreme states hold 0.41 each; the two middle states hold 0.09 each, far below the uniform 0.25 (dashed). New states carve up the rarely visited middle of the L axis, so they are rarely occupied and the alphabet goes underused.

(f) Boundaries go where time $\times$ usefulness peaks. Occupancy of the L axis (shaded), the usefulness $|dq/dL|$ of resolving there (red), and their product (black), against L_t . The $k = 4$ interval boundaries (grey dashed) sit at the peaks of the product, not of either factor alone. The belief spends most of its time near the stabilizing bound (high occupancy) but the posterior is already saturated there (low usefulness). The informative middle is visited rarely but changes fast. The optimizer balances both.

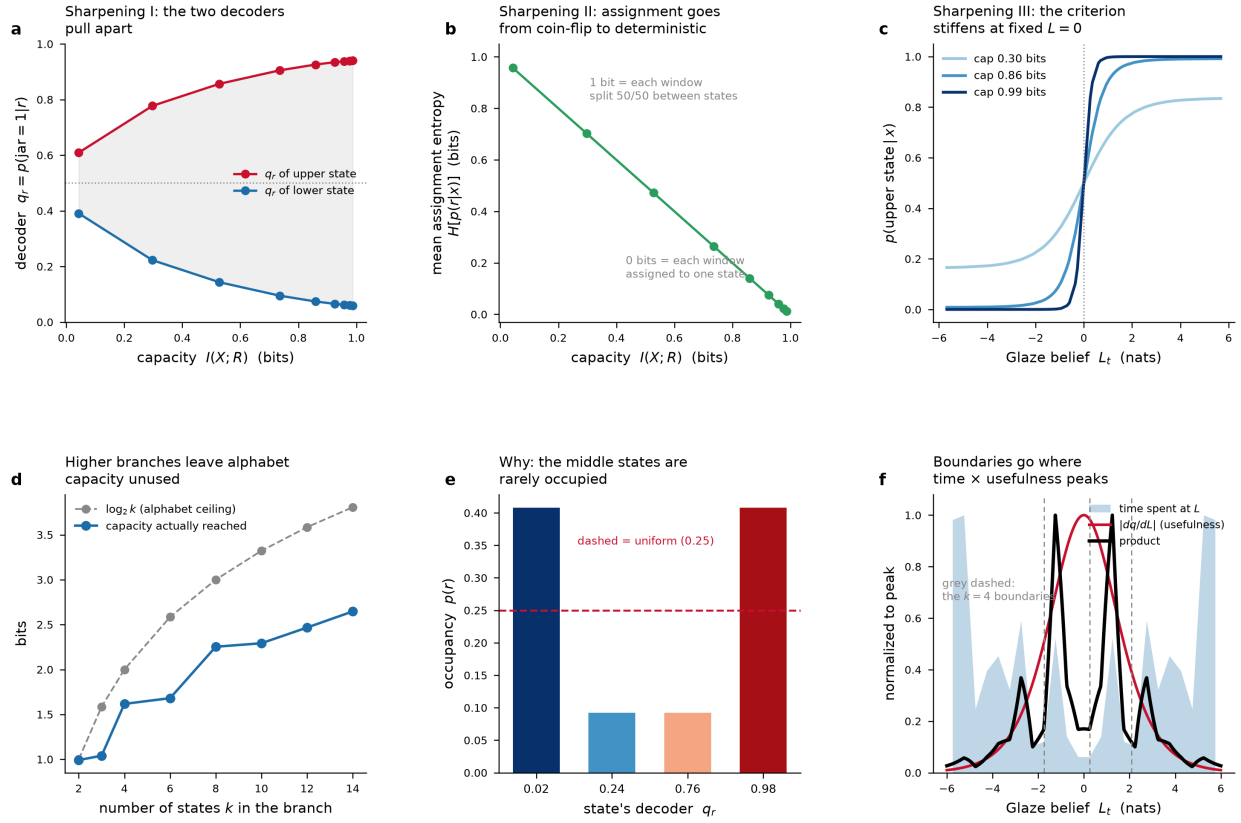


Figure 3: Sharpening within a branch, and the geometry that makes intervals contiguous

That is the whole mechanism: **sharpen, then split, then sharpen, then split**. It is why the frontier is a staircase and not a smooth curve.

2.2 Where the boundaries go, and why the branches are uneven

The intervals are not equal. The optimum places boundaries where they buy the most, and the rule is a product of two things:

Put a boundary where (how often the belief is there) $\times$ (how fast $p(\text{jar})$ changes there) is largest.

Both factors are strongly non-uniform, and they oppose each other. For the bead task at $h = 0.01$:

region of L	fraction of time spent there	mean $\backslash dq/dL $
$\backslash L < 1$	0.040	0.231
1–2	0.066	0.150
2–3	0.187	0.072
3–4	0.074	0.030
$\backslash L > 4$ (at the stabilizing bound)	0.633	0.009

The belief spends **63% of its time pinned near the stabilizing bound**, where the posterior is already saturated and additional resolution is nearly worthless. So the optimal code lumps each saturated tail into a single state and spends its remaining states carving up the middle — visible directly in Figure 2b–d, where boundaries cluster within roughly $|L| < 2$ and the tails are flat.

states k	capacity range (bits)	max $I(R; Y)$	branch top $\div \log_2 k$
2	threshold – 0.980	0.725	0.98
3	0.996 – 1.237	0.768	0.78
4	1.264 – 1.559	0.788	0.78
5	1.647 – 1.731	0.793	0.75
6	1.809 – 2.127	0.801	0.82

(belief space, $h = 0.01$, $|LLR| = \log 4$; ceiling $I(L; Y) = 0.806$ bits)

Only the two-state branch saturates. Beyond it, the new states carve up the rarely visited middle, so they are rarely occupied and contribute less capacity than a uniform alphabet would — the code stays unbalanced as it refines.

2.3 The first transition has a closed form

Start at very low β , where predictiveness is cheap relative to capacity. There the optimum is degenerate: one state, used for every window, decoder equal to the prior, zero capacity and zero predictiveness. The observer ignores the evidence. Now raise β and ask when that solution stops being optimal.

The δ in this argument is a **perturbation probe** — a bookkeeping device for a stability analysis, not a real quantity in the model. The logic is: start from the known trivial solution, hypothetically split it into two states whose decoders differ by a small amount δ , and ask whether the IB objective goes up or down. If it goes up for any $\delta > 0$, the trivial solution is unstable and the observer should split.

Expanding the objective in δ : the linear term vanishes (the trivial solution is a stationary point), and the quadratic term is what decides. Both the capacity cost and predictiveness gain of the split scale as δ^2 :

- Capacity cost of the split $\sim \delta^2 / [\bar{q}(1-\bar{q})]$
- Predictiveness gain $\sim \delta^2 \cdot \text{Var}[q] / [\bar{q}(1-\bar{q})]^2$

Because both scale identically in δ , the ratio gain/cost is independent of how large the split is. This means “does splitting pay off?” has the same answer for any $\delta > 0$ — either it always pays or it never does. δ cancels and the comparison reduces to a single ratio:

$$\beta_c = \bar{q}(1 - \bar{q}) / \text{Var}_{p(x)}[q(x)], \quad q(x) = p(y = 1 | x)$$

The interpretation is direct: the numerator is how uncertain the target is on average, the denominator is how much the evidence actually moves the posterior around. When the evidence barely moves q , β_c is large and the observer needs a high price on predictiveness before encoding anything at all.

The δ -cancellation is also the mathematical signature of a **continuous (second-order) transition**. If cost and benefit had scaled differently in δ — say cost $\sim \delta$ and benefit $\sim \delta^2$ — you would need to commit to a finite-size split before it paid off. That would produce a discontinuous (first-order) jump. The identical δ^2 scaling means the transition is marginal for any split size simultaneously, so capacity grows continuously from zero at β_c . The recursive transition in §7 has a different geometric structure and is abrupt (first-order) for exactly this reason.

For the bead task at $h = 0.01$, $W = 7$: **$\beta_c = 1.2326$** . Measured just above threshold, $I(X;R) \sim \epsilon^{0.95}$ and $I(R;Y) \sim \epsilon^{0.94}$ with $\epsilon = (\beta - \beta_c)/\beta_c$, both unity within solver tolerance, and the slope ratio $I(R;Y)/I(X;R) \rightarrow 0.8113$ against the predicted $1/\beta_c = 0.8113$ — at threshold each retained bit buys exactly $1/\beta_c$ bits of predictiveness. Validated independently against the optimizer’s own onset across seven hazards from $h = 0.002$ to 0.4 , maximum relative error 0.1% (Figure 1d).

2.4 The empirical prediction

Subjects in the manuscript sit almost entirely **below 1 bit** of capacity — which is the two-state branch, end to end. The prediction is therefore sharp and unusual:

A low-capacity observer does not hold a weak graded confidence. It holds a **binary belief** — “probably jar A” / “probably jar B” — separated by a criterion near $L = 0$, and what varies with capacity is how *sharp* that criterion is, not how graded the belief is.

The two-state branch tops out at $I(R;Y) = 0.725$ bits, against the manuscript’s reported 0.66-bit ideal-observer predictiveness — i.e. the reported human performance sits inside the range a strictly binary observer can reach. This is directly testable with confidence reports, and it separates the IB account from graded-belief accounts, which have no reason to predict a categorical structure at all.

Part II — Robustness of the quantization result

3. Continuous-valued observations do not dissolve it

The bead task delivers a binary observation per trial; the Glaze triangles task delivers a continuous one (a star’s horizontal position, drawn from a Gaussian centred on one of two triangles). A natural worry is that the quantization result is an artifact of the discrete observation alphabet, and that with continuous evidence the belief would be genuinely graded. It is not, and it would not.

Putting both tasks on one signal-to-noise axis. The SNR of the bead task is set by the proportion of the majority color in each jar, which in the manuscript is $p = 0.8$ (80% of one color in each jar). This single parameter determines everything downstream:

- One bead: $|\text{LLR}| = \log(p/(1-p)) = \log(0.8/0.2) = \log 4 \approx 1.386$ nats
- Log-odds separation between jars: $D = 2 \log 4$
- Equivalent Gaussian discriminability: $d' = \sqrt{D} = \sqrt{2 \log 4} \approx 1.66$

A different bead proportion — say $p = 0.7$ — would give $|\text{LLR}| = \log(7/3) \approx 0.85$ nats and $d' \approx 1.30$, a meaningfully weaker task. All specific numbers in this memo ($|\text{LLR}| = \log 4$, $d' = 1.66$) trace back to p

= 0.8. The bead task is therefore not a weak-evidence task — it is the triangles task at $d' = 1.66$, below Glaze’s two difficulty levels but not qualitatively apart from them:

condition	d'	ceiling $I(L; Y)$	β_c
beads, $p = 0.8$	1.66 (equivalent)	0.806	1.172
triangles, matched	1.66	0.862	1.112
triangles, Glaze mid ($\sigma/\text{distance} = 0.33$)	3.03	0.963	1.027
triangles, Glaze easy ($\sigma/\text{distance} = 0.24$)	4.17	0.989	1.008

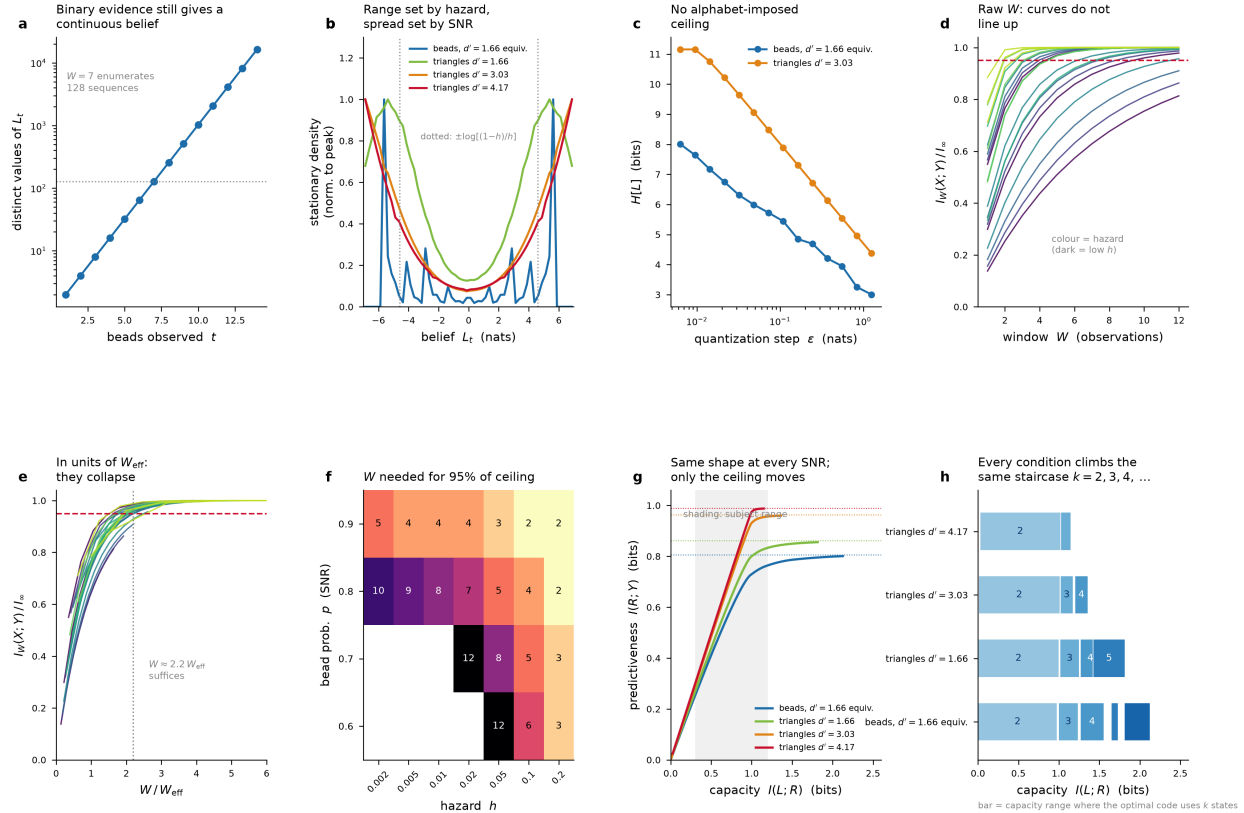


Figure 4: Beads and triangles under the information bottleneck

Figure 4. The window W is not a capacity ceiling, and the bottleneck acts the same way on binary and continuous evidence. All panels $h = 0.01$ unless stated.

(a) Number of distinct values the belief L_t can take after t beads, computed exactly by propagating the reachable set (merging at $1e-12$, no grid). The count is 2^t : because the discount ψ is nonlinear, $\pm \log 4$ increments never re-land on a shared value. Verified to $t = 20$ (1,048,576 distinct values, no collapse). The nonlinearity of ψ means each new path through belief space is unique. The $W=7$ window is not discretizing a continuous belief — it is limiting the history conditioned on. Doubling holds for all $h < 0.5$. Dotted line marks the 128 sequences a $W = 7$ window enumerates.

(b) Stationary distribution of the belief for all four conditions, binned at 0.25 nats and normalized to peak. All four conditions share the same stabilizing bound $\pm \log [(1-h)/h] = \pm 4.60$ nats (dotted lines), set by hazard alone — not by SNR. What d' changes is how mass distributes *within* that range. For strong evidence ($d'=4.17$), mass concentrates more sharply at the extremes — reinforcing the case for a coarse code. All four are bounded in range but continuous in value.

(c) Entropy of the belief quantized at resolution ϵ . Both tasks grow without bound as $\epsilon \rightarrow 0$; neither has a capacity ceiling set by its observation alphabet. The discretization in Figure 2 was never inherited from the stimulus — it is entirely imposed by the capacity limit.

(d) Fraction of the unbounded-memory evidence ceiling that a window of size W recovers at 1 bit of capacity, for hazards 0.002–0.2 (colour, dark = low hazard) at three bead probabilities. Plotted against raw W the curves do not line up: the window a task needs depends on both hazard and SNR.

(e) The same curves against W / W_{eff} , where $W_{\text{eff}} = \log[(1-h)/h] / E[|LLR|]$ is the number of observations needed to traverse the stabilizing bound. All curves collapse, and $W \approx 2.2 W_{\text{eff}}$ (dotted) suffices to reach 95% of the ceiling (dashed). For the manuscript's main condition ($p=0.8$, $h=0.01$): stabilizing bound ≈ 4.60 nats, $E[|LLR|] = \log 4 \approx 1.386$ nats/bead, $W_{\text{eff}} \approx 3.3$, so $W \approx 7.3$ — confirming $W=7$ is well-matched to this condition.

(f) The window required for 95% of the ceiling, as a grid across hazard rate h and bead probability p . Largest requirements cluster at low h and low p . For the manuscript's condition ($p=0.8$, $h=0.01$), $W=7$ is acceptable (92.6%). Blank cells: $W=12$ does not suffice.

(g) IB frontiers for all four conditions, with X = the belief itself. The shape is identical across conditions — same staircase, same branch pattern. Only the evidence ceiling (dotted, per condition) changes. SNR lifts or lowers the curve but does not change its shape. Shading marks the capacity range where subjects sit — below ~ 1 bit for most — which is the same relative position on all four frontiers.

(h) Branch map: for each condition, the capacity range over which the optimal code uses k states. Every condition climbs the same staircase, and the onsets barely move with SNR. Across a $2.5\times$ range of d' , $k=3$ first appears at 1.00–1.02 bits, $k=4$ at 1.20–1.26 bits. **The number of categories a subject holds is a statement about their capacity, not about the difficulty of the stimulus.** SNR determines how much predictiveness each state buys; the staircase rungs themselves are fixed by capacity alone. Shading marks the subject range.

Three findings matter here.

The bead-task belief is already continuous. Because ψ is nonlinear, successive $\pm \log 4$ increments never coincide: the reachable set of L_t is exactly 2^t , verified to $t = 20$ (1,048,576 distinct values, no collapse at merge tolerance $1e-12$) and confirmed for every $h \in [0.002, 0.4]$. A binary observation alphabet does not produce a discrete belief. So the discreteness in Figure 2 was never inherited from the stimulus; it is imposed by the capacity limit.

The cascade survives genuinely continuous evidence. Running the bottleneck with X = the belief itself, on Gaussian observations at the Glaze signal levels, reproduces the same staircase:

task ($h = 0.01$)	ceiling $I(L;Y)$	2-state branch tops at	$I(R;Y)$ there	fraction of ceiling
beads, $ LLR = \log 4$	0.806	0.980 bits	0.725	90.0%
triangles, $d' = 1.66$ (matched)	0.862	1.002	0.800	92.8%
triangles, $d' = 3.03$ (Glaze mid)	0.963	1.014	0.935	97.1%
triangles, $d' = 4.17$ (Glaze easy)	0.989	1.022	0.981	99.2%

Continuous observations do not buy graded belief for free. Note also that the two-state branch captures a *larger* fraction of the ceiling as evidence strengthens, which makes the categorical prediction of §2.4 easier to satisfy, not harder.

SNR sets the ceiling; capacity sets the number of states. These are separable. Across a 2.5× range of d' , the capacity at which each new state appears barely moves:

condition	k = 2 from	k = 3 from	k = 4 from
beads ($d' = 1.66$ equiv.)	threshold	1.00	1.26
triangles $d' = 1.66$	threshold	1.01	1.26
triangles $d' = 3.03$	threshold	1.02	1.20
triangles $d' = 4.17$	threshold	1.02	—

The number of categories a subject holds is a statement about their capacity, not about the difficulty of the stimulus.

4. The window W is a numerical device, not a capacity ceiling

The manuscript estimates $p(x, y)$ by enumerating sequences over a finite window W . It is worth being explicit that this is a tractability device for estimating the empirical joint distribution and never a claim about subject memory — windowed and unrestricted Bayesian observers agree on 98.6% of predictions as reported, and 98.3% by direct simulation ($h = 0.01$, $W = 7$).

The windowed variable has $H(X) = W$ bits, so the windowed problem does have a formal ceiling of W bits. But the IB frontier saturates at $I(X;Y)$ instead, and $I(X;Y)$ is smaller by an order of magnitude:

All rows at $h = 0.01$, $|LLR| = \log 4$:

W	$H(X) = \text{capacity } W \text{ allows}$	$I(X;Y) = \text{capacity worth using}$	$I(R;Y) \text{ at 1 bit}$
3	3 bits	0.571	0.494
5	5	0.702	0.615
7	7	0.760	0.674
10	10	0.792	0.709
12	12	0.800	0.718
∞ (belief space)	∞	0.806	0.725

At this hazard the first column never binds: the $W = 7$ and $W = \infty$ frontiers are indistinguishable below ~ 1.5 bits, and subjects sit below 1 bit. Every capacity range quoted in this memo is therefore a statement about **belief resolution**, not window size; re-solving with no window at all changes them by $<2\%$. The range of L is set by the hazard's stabilizing bound (Figure 4b), not by W .

5. Hazard dependence, including where W does bind

β as a price per bit of capacity

Before reading the hazard-dependence table, it helps to have a concrete intuition for what β means. Rearrange the IB objective to make the economics explicit:

$$\text{maximize: } \beta \cdot I(R;Y) - I(X;R)$$

β is the **exchange rate** between predictiveness and capacity — how many bits of predictiveness justify spending one bit of capacity. If $\beta=2$, you will spend up to 2 bits of capacity to gain 1 bit of predictiveness. If $\beta=10$, you will spend 10 bits for that same gain.

β_c is the **minimum viable exchange rate** — the lowest β at which drawing even the first boundary pays off. Below β_c , no split ever earns back its capacity cost. Above it, any split pays — even an infinitesimally small one earns more predictiveness than it costs in capacity.

Critically, β_c is a property of the **task**, not of the observer. It depends on how much the evidence moves the posterior ($\text{Var}[q]$) and how uncertain the target is on average ($\bar{q}(1-\bar{q})$). An observer with a fixed β will engage with evidence in tasks where $\beta > \beta_c$ and ignore it in tasks where $\beta < \beta_c$. This makes β_c the key parameter for predicting when a capacity-limited observer should disengage from evidence encoding entirely.

The ceiling falls steeply with hazard — a faster-switching source is less predictable — and β_c rises with it. Belief space, beads, $|\text{LLR}| = \log 4$:

h	stabilizing bound (nats)	$I(X;Y)$ (bits)	β_c
0.002	6.213	0.931	1.048
0.005	5.293	0.873	1.100
0.01	4.595	0.806	1.172
0.02	3.892	0.717	1.292
0.05	2.944	0.571	1.567
0.10	2.197	0.455	1.899
0.20	1.386	0.353	2.334
0.30	0.847	0.307	2.591

In words: when the world switches often, evidence about the current state is worth less, so a capacity-limited observer requires a larger β — a lower effective price per bit — before encoding anything at all. At $h = 0.3$ predictiveness must be valued 2.6× the bit price; at $h = 0.002$, barely 1.05×.

w binds at low hazard. The fraction of the $w = \infty$ frontier a window recovers, at 1 bit of capacity:

h	$w=3$	$w=7$	$w=12$
0.002	0.573	0.852	0.958
0.01	0.678	0.926	0.988
0.05	0.881	0.991	1.000
0.20	0.996	1.000	1.000

$w = 7$ is adequate at $h \geq 0.05$ (>99%) and acceptable at $h = 0.01$ (92.6%), but **not at $h = 0.002$ (85.2%)**. The requirement is set by w_{eff} , the number of observations needed to traverse the stabilizing bound (Figure 4d–f): windows collapse onto a single curve in units of w_{eff} , and $w \approx 2.2 w_{\text{eff}}$ suffices for 95% of the ceiling. So w is never a capacity ceiling, but the claim that it is *non-binding* holds only for $h \geq 0.01$; below that, use the belief-space solve.

A dissociation the design can exploit. β_c rises with h in both tasks, but at very different rates:

h	β_c beads	β_c triangles ($d' = 3.03$)
0.002	1.048	1.008
0.01	1.172	1.027
0.05	1.567	1.081
0.10	1.899	1.124
0.20	2.334	1.179
0.30	2.591	1.213

A 2.5× swing against a 1.2× swing — a genuine $h \times d'$ interaction, not two main effects. A capacity-limited observer's threshold for engaging with the evidence at all should be strongly volatility-sensitive when evidence is weak and nearly volatility-insensitive when it is strong. A second, separable prediction concerns instruction: with $Y = \text{next jar}$ rather than current jar, $\beta_c^{\text{next}} = \beta_c^{\text{now}} / (1 - 2h)^2$ exactly (1.283

at $h = 0.01$ rising to 68.3 at $h = 0.40$). Prediction becomes catastrophically harder to justify representing as volatility rises, while filtering degrades gently.

Range caution: Glaze's blocks run to $h = 0.95$. Everything above is $h \leq 0.45$. At $h = 0.5$ the discount vanishes identically; above 0.5 the source alternates more often than it persists, inverting the sign of useful evidence. That regime needs the theory restated, not merely re-run.

Part III — Recursion

6. Requiring the belief to be updated from itself is costly, and costly exactly where subjects are

Everything above solves a **one-shot** problem: R is computed from the evidence in a single step, with no requirement that it be updatable from its own previous value. Real accumulation is **recursive**: $r_t = f(r_{t-1}, x_t)$, one new observation at a time. Whether that restriction costs anything is the central open question of the project, and it does.

How a recursive accumulator appears on these axes. A recursive machine is not an IB solution — it is a mechanism, and it gets *scored* on the same two axes. Specify k states and a transition rule $r_t = f(r_{t-1}, x_t)$; run that chain to its stationary distribution over (r_t, y_t) ; read off $I(R;Y)$ for the vertical axis and the capacity the machine consumes for the horizontal one. Each machine is one point, and the frontier is the upper envelope of the point cloud.

The horizontal axis requires care. For a **deterministic** machine, r_t is a function of the entire past, so $I(x_{1:t}; r_t) = H(R)$ and that entropy is directly comparable to the one-shot $I(X;R)$. For a **stochastic** kernel the identity fails — $H(R)$ then also counts the noise the kernel injects, which is not information about the stimulus — so only deterministic machines are plotted and only they enter the envelope.

Figure 5. Recursion is costly at low capacity, and its optimum is a bounded ladder. Bead task, $h = 0.01$; one-shot reference is $W = 12$ ($I(X;Y) = 0.7997$), since the recursive machine sees unbounded history.

(a) The one-shot IB bound (blue) against every deterministic recursive machine found (grey points, $k \leq 8$) and their upper envelope (orange). The shaded region between the two curves is the **cost of recursion** — predictiveness lost by requiring the observer to update from its own previous state rather than recomputing from the evidence afresh. This cost is 13–38% of the one-shot bound in the subject range (below ~1 bit) and shrinks to zero by ~2.25 bits. Dotted line is the unbounded-memory ceiling. Grey band is the capacity range where subjects sit.

(b) Ratio of the recursive envelope to the one-shot bound — what a *perfectly efficient* recursive observer scores when graded against the one-shot bound. This is a **prediction**, not a measurement: even a subject who is perfectly optimal given recursive constraints will appear 62–87% efficient against the one-shot bound in the subject range. The apparent inefficiency in the low-capacity tail may be the signature of recursive inference rather than of suboptimality. Testable by refitting subjects against the recursive bound on the existing dataset.

(c) The optimal 4-state machine as a state diagram. Nodes are the four states (ordered left to right), with decoder values $p(\text{jar}=1|\text{state}) = 0.02 / 0.23 / 0.77 / 0.98$ shown below. A green bead moves the state one rung up; a red bead moves it one rung down; the extreme states are non-absorbing. This is a **bounded ladder**: a leaky accumulator with non-absorbing bounds (e.g., from Glaze et al.) but here in discrete form. Here it falls out of exhaustive search over all 65,536 possible wirings of 4 states and 2 bead colors, with no structural assumptions. The decoder values are ordered but not equally spaced: outer states are more extreme, covering the tails where the belief spends most of its time; inner states cover the informative middle. Note that the optimal 4-state recursive accumulator has the same architectural structure as the Glaze leaky accumulator — ordered states, evidence-driven updates, non-absorbing bounds — and can be understood as the optimal discrete approximation to the Glaze filter under a capacity constraint. However,

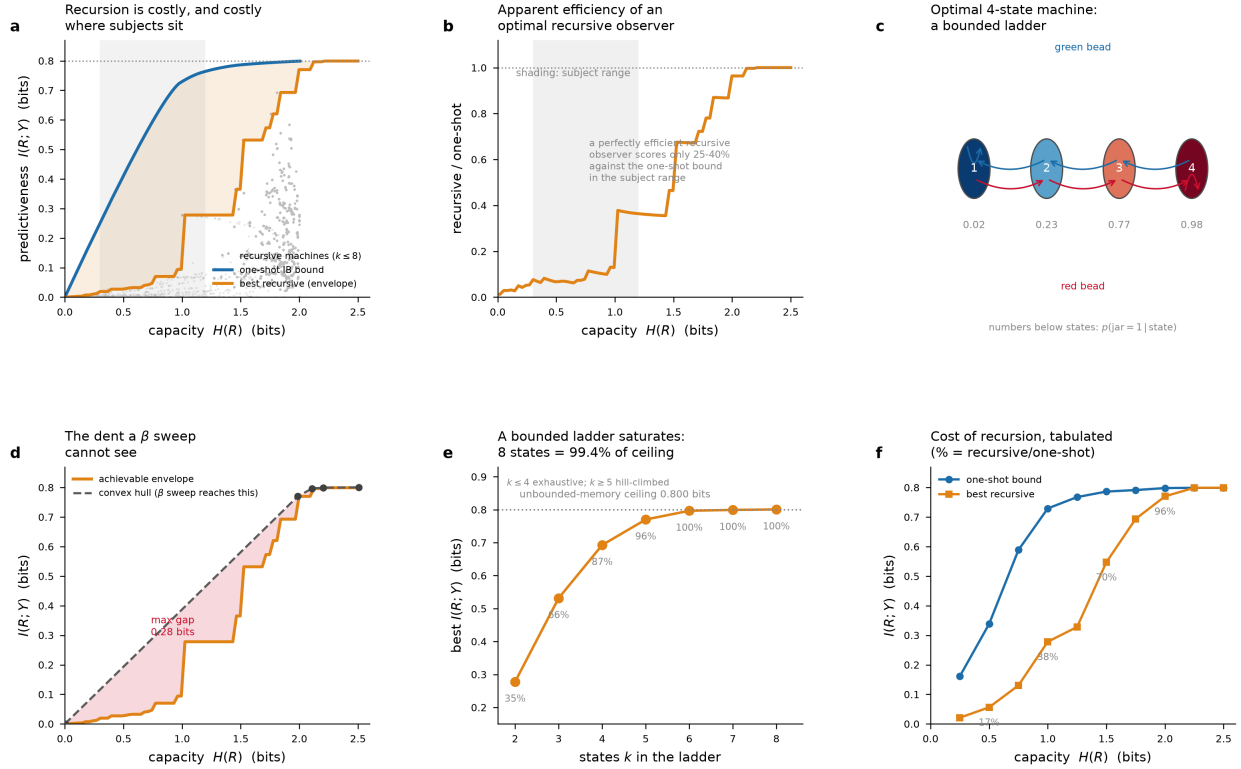


Figure 5: Cost of recursion and the optimal recursive accumulator

the transition dynamics differ: the Glaze model applies a continuous, L -dependent leak that compresses extreme beliefs back toward the stabilizing amplitude before each update, while the IB discrete model applies a uniform ± 1 step regardless of position within a state. The two converge in the limit of large k (many discrete states) but diverge substantially for extreme states at $k = 4$.

(d) The achievable envelope (orange) against the upper convex hull of the same point cloud. A β sweep can only select points on the upper convex hull. If the achievable region has an inward dent, the β sweep jumps across it — no value of β selects anything inside. The recursive achievable region has such a dent: the hull exceeds the achievable envelope by up to 0.28 bits, centered around 0.92 bits, across 0.05–2.07 bits. Consequences: (1) the recursive frontier must be computed by constrained search at fixed capacity, not by sweeping β — a β sweep silently overstates what recursion can do; (2) the transition is abrupt (§7); (3) the dent sits in the subject range.

(e) Best $I(R;Y)$ against the number of states k in the ladder. $k \leq 4$ exhaustively enumerated; $k \geq 5$ hill-climbed (lower bounds). $k=2$: 25% of ceiling; $k=4$: 86%; $k=6$: 100%; $k=8$: 99.4%. Returns diminish quickly — a 6-state ladder is essentially perfect. Bounded, discrete accumulation is nearly lossless when capacity permits.

(f) The cost-of-recursion table (§6 below), plotted, with the ratio of recursive to one-shot annotated at four capacities. Blue is the one-shot bound; orange is the best recursive envelope. Given a subject's estimated capacity $H(R)$, the orange curve gives the predictiveness a perfectly efficient recursive observer should achieve at that capacity — suitable for fitting to subject data.

The optimal recursive accumulator is a bounded ladder. Exhaustive search over all 65,536 wirings of 4 states and 2 bead colours returns next-state[state,bead] = $[[0,1], [0,2], [1,3], [2,3]]$ with $p(\text{jar}=1|\text{state}) = [0.02, 0.23, 0.77, 0.98]$. This is precisely the architecture Glaze et al. derive analytically — a leaky accumulator with non-absorbing bounds — recovered here as the solution to an information-constrained optimization rather than assumed. The optimal $k = 2$ machine ignores its own

previous state entirely, making it exactly the one-back heuristic the manuscript already fits: the one-back rule is not a shortcut but the *optimal* recursive strategy at 1 bit. An 8-state ladder reaches 0.8007 bits, **99.4% of the unbounded-memory ceiling** — bounded, discrete accumulation is nearly lossless when capacity permits.

The cost is concentrated below 2 bits. Envelope over all 34,564 deterministic machines:

capacity (bits)	one-shot bound	best recursive	ratio
0.25	0.161	0.021	13%
0.50	0.339	0.056	17%
0.75	0.589	0.130	22%
1.00	0.729	0.278	38%
1.25	0.768	0.328	43%
1.50	0.786	0.547	70%
1.75	0.792	0.694	88%
2.00	0.798	0.770	96%
2.25	0.799	0.799	100%

At high capacity recursion is free; in the subject range a recursive accumulator reaches **13–38%** of what an unconstrained one-shot code achieves at the same capacity.

This reinterprets the manuscript's efficiency measure. Subjects are scored against the one-shot bound, and most sit below 1 bit — exactly where the two bounds diverge by a factor of three or more. A subject who is *perfectly efficient given that their inference must be recursive* will look substantially *inefficient* against the one-shot bound. The apparent inefficiency in the low-capacity tail may be the signature of recursion rather than of suboptimality, and this is testable on the existing dataset by refitting the same subjects against the recursive bound.

7. The recursive transition is abrupt

The one-shot transition of §2.3 is continuous: capacity grows smoothly from zero at β_c . The recursive problem does not behave that way, and the reason is geometric.

A β sweep optimizes $I(R;Y) - (1/\beta) \cdot I(X;R)$, which is a linear objective on the (capacity, predictiveness) plane. A linear objective can only ever select points on the **upper convex hull** of the achievable region. If the region is convex, the hull and the region's upper boundary coincide and the sweep finds everything. If the region has an inward dent, the sweep jumps across it: as β increases, the optimum leaps discontinuously from one side of the dent to the other, and no value of β selects anything inside.

The bead task's recursive region has such a dent. Taking all 34,564 deterministic machines, constructing their upper convex hull, and comparing it against the achievable envelope (Figure 5d): the hull exceeds the envelope by up to **0.28 bits**, at 0.92 bits of capacity, with a gap above 0.01 bits across the range 0.05–2.07 bits. Good machines exist throughout that range — they are simply unreachable by any β .

That is the signature of an **abrupt (first-order)** transition: the optimal strategy switches discontinuously rather than deforming smoothly. The one-shot and recursive problems therefore change character in fundamentally different ways, and the difference is not an artifact of the search — it is a property of which (capacity, predictiveness) pairs finite recursive machines can occupy at all.

Two practical consequences. First, any recursive frontier must be computed by constrained search at fixed capacity, never by sweeping β ; a β sweep silently returns the hull and overstates what recursion can do in the dent. Second, the dent sits squarely in the subject range, which is where §6's reinterpretation of the efficiency measure lives.

8. Methods

Windowed problem. For window w , enumerate all 2^w bead sequences; compute $p(x)$ and $p(y|x)$ exactly by forward recursion over the HMM (no sampling). Merge sequences with identical $p(y|x)$ — they are interchangeable for any IB solution. Solve $\min I(X;R) - \beta I(R;Y)$ by Blahut–Arimoto with restarts, swept over β . Y is the **current source jar** throughout; the manuscript’s reported ideal-observer values (0.66 bits, 93.9% accuracy) identify it as such, against 0.760 bits / 94.1% for current jar, 0.712 / 93.2% for next jar, and 0.214 / 75.9% for next bead colour.

Belief-space problem (no window). Build the stationary joint $p(L_t, y_t)$ for the Glaze recursion on a grid of L ($n = 1201$ – 2001 nodes spanning the stabilizing bound plus one observation’s evidence). Binary evidence splits probability mass linearly between neighbouring nodes; Gaussian evidence uses CDF differences. The kernel is asserted exactly stochastic, then iterated to its stationary distribution. Validation: the bead chain’s $I(L;Y)$ reproduces the independently computed unbounded-memory ceiling 0.8058 bits, and $p(y=1|L)$ matches $\sigma(L)$ to machine precision.

SNR matching. One bead carries $|\text{LLR}| = \log 4$, giving a log-odds separation $D = 2 \log 4$ between jars; the Gaussian observation with the same separation has $d' = \sqrt{D} = 1.66$. Glaze’s $\sigma/\text{distance}$ ratios 0.24 and 0.33 correspond to $d' = 4.17$ and 3.03.

Support growth (Figure 4a) is exact: the reachable set of L_t is propagated atom-by-atom with merging at $1e-12$, no grid.

State counting. Solutions are collapsed on $q = p(y=1|L)$, since the IB depends on x only through q . Counting states by an absolute separation threshold is unreliable near β_c , where every decoder converges on the prior and numerically split copies of one state look distinct — such a rule reports three states at 0.001 bits of capacity, which $I(L;R) \leq \log_2 k$ forbids. States are therefore counted by an information criterion instead: repeatedly merge the two closest decoders (mass-weighted) while the merge costs less than 1% of $I(R;Y)$, and stop when it does not. The reported k is the number of states carrying distinguishable information, and it is scale-free — no absolute tolerance enters.

Recursive problem. Stationary joint $p(r_t, y_t)$ as the leading eigenvector of the induced Markov chain; capacity $H(R)$, predictiveness $I(R; y_t)$. Validated against 3×10^6 -step simulation (3–4 decimals). Deterministic machines exhaustively enumerated for $k \leq 4$ (34,136 machines at $k = 4$), hill-climbed with hundreds of restarts for $k \leq 8$; stochastic kernels by Nelder–Mead seeded from the best deterministic machines. Frontiers, envelopes and the convex-hull test of §7 use **deterministic machines only**, since $I(x_{1:t}; r_t) = H(R)$ holds only for them; mixing machine classes on this axis is not meaningful. The one-shot reference is $W = 12$ ($I(X;Y) = 0.7997$), not $W = 7$ (0.7598), because the recursive machine sees unbounded history.

9. What this buys the grant

1. **A mechanism for categorical belief.** Capacity limits do not blur the normative belief; they quantize it (§2). The prediction that low-capacity subjects hold binary belief with a sharp criterion — not weak graded confidence — is testable with confidence reports and has no counterpart in graded-belief accounts.
2. **Two knobs that do different jobs.** SNR sets the ceiling and β_c ; capacity sets the number of categories (§3). Across a $2.5\times$ range of d' the state-count onsets move by less than 0.06 bits, so the number of categories a subject holds is diagnostic of their capacity rather than of task difficulty.
3. **The normative model explained rather than replaced.** Glaze’s leaky accumulator with non-absorbing bounds falls out of information-constrained recursive inference (§6) instead of being assumed.
4. **A separable two-parameter prediction grid.** β_c falls with d' and rises with h , with a strong interaction between them (§5), plus a parameter-free instruction effect $\beta_c^{\text{next}}/\beta_c^{\text{now}} = 1/(1-2h)^2$. The existing Glaze dataset (48 subjects, 1000-trial blocks, three d' levels) already spans it.

5. **A reinterpretation of the manuscript's efficiency measure.** §6 predicts that low-capacity subjects appear inefficient against the one-shot bound even when optimal, and quantifies by how much — testable on existing data.
6. **Two transitions of different order** (§2.3, §7): the one-shot transition is continuous with a closed-form threshold, the recursive one is abrupt, and the difference is a property of the achievable region rather than of the search.

10. Open items

- Does the §7 dent survive $k > 8$? The hull test is exact for the machines enumerated, and $k \leq 4$ is exhaustive, but $k \geq 5$ rests on hill-climbing.
- Can the recursive threshold be derived analytically, to match §2.3's closed form?
- Refit the manuscript's subjects against the recursive bound (§6) — needs raw choice data.
- Extend the theory above $h = 0.5$ to cover Glaze's high-hazard blocks (§5).
- Confidence-report design to test the categorical prediction of §2.4 directly.
- Continuous-state accumulators (the Glaze diffusion limit) rather than finite k .
- Reconcile the two-state branch top onto a single β grid: 0.9808 bits in `belief_space_summary.csv` versus 0.9916 in `branch_alphabet_usage.csv` (§2).

11. Files

Regenerated from the artifact store for v5; entries that no longer resolve have been removed and previously undocumented outputs added.

Code

- `ib_core.py` — computational core: windowed HMM enumeration, exact collapse, Blahut–Arimoto, closed-form β_c , deterministic and stochastic recursive solvers, simulation validators.

Belief-space sweeps and the branch map (Parts I–II)

- `belief_space_ib_sweep.csv` — belief-space IB sweeps, all four task variants ($h = 0.01$).
- `branch_map_belief.csv` — state count versus capacity for all four conditions, under the §8 counting rule (Figure 4h, §3 table).
- `belief_space_summary.csv` — per-task ceilings, β_c , two-state branch extent.
- `sharpening_within_branch.csv` — decoder separation, mean assignment entropy and mean confidence across the two-state branch (Figure 3a–c).
- `branch_alphabet_usage.csv` — capacity reached at the top of each k -state branch against the $\log_2 k$ alphabet ceiling, $k = 2\text{--}14$ (Figure 3d).

Window adequacy and capacity ceilings

- `window_vs_belief_frontier.csv` — windowed frontiers, $w = 3, 5, 7, 10, 12$.
- `w_adequacy_hazard_snr.csv`, `w_eff_requirement.csv` — window adequacy across hazard and SNR, and the w_{eff} collapse (Figure 4d–f, §5).
- `capacity_ceiling_comparison.csv` — $H(X)$ vs $I(X;Y)$ (§4).
- `ib_bound_w12.csv` — $w = 12$ reference bound.

Hazard dependence (§5)

- `hazard_dependence.csv`, `beta_c_vs_hazard.csv`, `hazard_frontier_at_1bit.csv`.
- `ib_leak_hazard_graded.csv` — optimal leak λ^* and its within-tolerance band against hazard, with the Glaze $1 - 2h$ comparison.

One-shot and accumulation analyses

- `ib_bead_jar_sweep.csv` — one-shot bead-jar sweep.

- `ib_accumulation_results.csv` — IB maximum against normative and perfect-integration bits across h and T , with efficiency ratios.
- `ib_capacity_leak_tradeoff.csv` — best leak and its tolerance band as a function of capacity budget.
- `ib_dynamic_bottleneck_sweep.csv` — dynamic-bottleneck sweep over β ($I(X;R)$, $I(R;Y)$, cluster count, $I(R;L)$).

Recursion (Part III)

- `recursive_points_all.csv` — all recursive machines found.
- `recursive_envelope_true.csv` — achievable envelope over deterministic machines (Figure 5a,d).
- `recursive_constrained_frontier.csv` — constrained frontier by k and capacity.
- `cost_of_recursion_corrected.csv` — the §6 table.

Notes

- `memo_QA_log.md` — audit log; §Q1b records the withdrawn sufficiency ratio and §Q2 the exact provenance of Figure 1.
- `lab_corpus_index.csv` — index of the lab reference corpus (251 entries); used for citation lookup, not an analysis output.

Removed from this list (no longer in the artifact store)

- `belief_ib_solutions.pkl` — stored Blahut–Arimoto solutions behind the branch map. Regenerate with `ib_core.ba` if the branch map needs rebuilding.
- `one_shot_frontier_beads.csv` — superseded by `ib_bead_jar_sweep.csv`.